6. Hardening

The plantlets developed *in vitro* require a hardening period and so are transferred to greenhouse or hardening chamber and then to normal environmental conditions.

Hardening is the gradual exposure of *in vitro* developed plantlets in humid chambers in diffused light for acclimatization so as to enable them to grow under normal field conditions.

5.3.3 Types of Plant tissue cultures

Based on the type of explant other plant tissue culture types are

1. Organ culture
2. Meristem culture
3. Protoplast culture
4. Cell suspension culture.

1. Organ culture

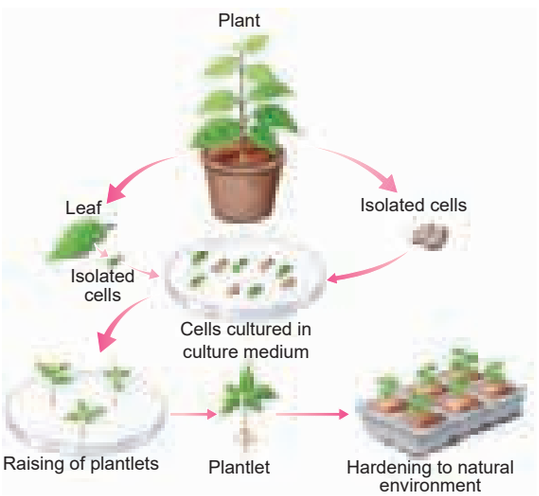


Figure 5.6: Organ Culture

The culture of embryos, anthers, ovaries, roots, shoots or other organs of plants on culture media.

2. Meristem Culture:

The culture of any plant meristematic tissue on culture media.

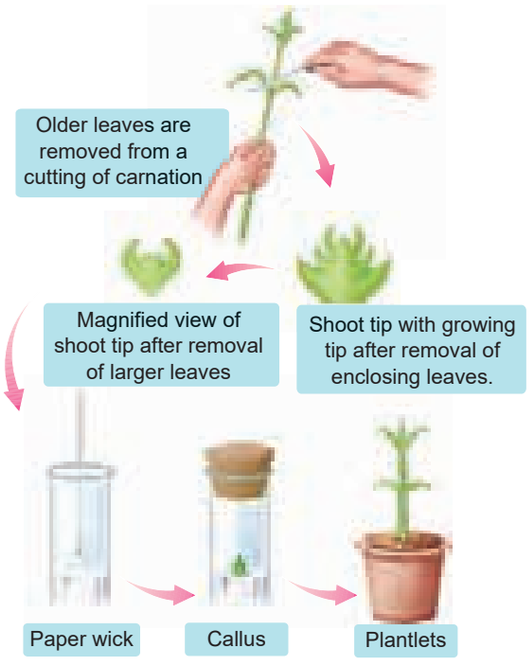


Figure 5.7: Meristem Culture

3. Protoplast Culture:

Protoplasts are cells without a cell wall, but bound by a cell membrane or plasma membrane. Using protoplasts, it is possible to regenerate whole plants from single cells and also develop somatic hybrids. The steps involved in protoplast culture.

i. **Isolation of protoplast:** Small bits of plant tissue like leaf tissue are used for isolation of protoplast. The leaf tissue is immersed in 0.5% Macrozyme and 2% Onozuka cellulase enzymes dissolved in 13% sorbitol or mannitol at pH 5.4. It is then incubated over-night at 25°C. After a gentle teasing of cells, protoplasts are obtained, and these are then transferred to 20% sucrose solution to retain their viability. They are then centrifuged to get pure protoplasts as different from debris of cell walls.

ii. **Fusion of protoplast:** It is done through the use of a suitable fusogen. This is normally PEG (Polyethylene Glycol). The isolated protoplast are incubated in 25 to 30% concentration of PEG with Ca++ ions and the protoplast shows agglutination (the formation of clumps of cells) and fusion.

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